

Wendy Q. Sun

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EDUCATION

Massachusetts Institute of Technology

Cambridge, MA

Double major: B.S. in Physics

2022 - 2026

B.S. in Artificial Intelligence & Decision Making

Minor in Mathematics

GPA : 4.9/5.0

HONORS

Phi Beta Kappa Honor Society (top 10% of the graduating class)

2026

Sigma Pi Sigma Physics and Astronomy Honor Society

2026

Tau Beta Pi Engineering Honor Society

2026

PUBLICATIONS

3 primary author papers, [ADS library](#). 16 total papers, [ADS library](#).

[†] marks equal contribution.

Primary Author Papers

3. **W. Q. Sun**, R. P. Naidu, H. Liu, A. de Graaff, J. E. Greene, J. Matthee, et al., *Overmassive No More: The Case for Little Red Dots Hosting Black Hole Seeds as Massive as Single Supermassive Stars*, [submitted to OJAp](#).
2. **W. Q. Sun**, R. P. Naidu, J. Matthee, A. de Graaff, et al., *Little Red Dot – Host Galaxy = Black Hole Star: A Gas-Enshrouded Heart at the Center of Every Little Red Dot*, [OJAp](#), **9**, 62505, 2026.
1. C. Loughridge[†], **Q. Sun**[†], et al., *DafnyBench: A Benchmark for Formal Software Verification*, [The 4th Workshop on Mathematical Reasoning and AI at NeurIPS, 2024](#).

Contributing Author Papers

13. G. Limberg et al., *Chemodynamical Analysis of a CNO-Enhanced Ultra Metal-poor Star ($[Fe/H] < -4$): Insights into Early Enrichment by Faint Population III Supernovae*, [submitted to AAS journals](#).
12. C. Ashall et al., *Can Little Red Dots Contribute To The Early Universe Cosmic Dust Budget?*, [submitted to ApJ](#).
11. R. P. Naidu et al., *A gas-enshrouded and gas-reddened black hole at cosmic dawn*, [Nature](#), **656**, 329, 2026.
10. Y. Ishikawa et al., *Spatial decomposition of Little Red Dots with JWST/NIRSpec IFU into broad-line red cores and narrow-line blue host galaxies*, [arXiv](#).
9. R. P. Naidu et al., *Little Red Dots as Intermediate Mass, Super-Eddington Engines: Insights from Type II_n Supernovae and The 1837–1856 Great Eruption of η Carinae*, [arXiv](#).
8. A. Weibel, R. P. Naidu, et al., *Black Hole Stars Across the Universe: Identifying Central Engine Dominated Little Red Dots at $z \sim 1.5 - 9.5$* , [arXiv](#).
7. B. Wang et al., *Water absorption confirms cool atmospheres in two little red dots*, [arXiv](#).
6. A. Torralba et al., *A Black Hole Star at Cosmic Noon: Extreme Balmer break, photospheric continuum, and broad absorption by thick winds in a Little Red Dot at $z = 1.7$* , [ApJL](#), **1005**, 37, 2026.
5. Z. Liu, R. P. Naidu, A. Secunda, et al., *How I Wonder What You Are – JWST’s Little Red Dots do not TWINKLE*, [submitted to ApJ](#).
4. T. Li et al., *Fractal Generative Models*, [Transactions of Machine Learning Research \(TMLR\)](#), 2025.

3. J. C. Carr et al., *Focused Skill Discovery: Learning to Control Specific State Variables while Minimizing Side Effects*, [The 2nd Reinforcement Learning Conference \(RLC\)](#), 2025.

2. S. Casper, C. Ezell, et al., *Black-Box Access is Insufficient for Rigorous AI Audits*, [The ACM Conference on Fairness, Accountability, and Transparency \(FAccT\)](#), 2024.

1. W. Gurnee et al. *Universal Neurons in GPT2 Language Models*, [Transactions of Machine Learning Research \(TMLR\)](#), 2024.

PRESENTATIONS

ISSI Bern, *LRD – Host Galaxy = BH*: A Gas-Enshrouded Heart at the Center of Every LRD*
Meeting, Feb. 2026

RESEARCH EXPERIENCE

Weighing Central Engines of Little Red Dots

Honolulu, HI

Institute for Astronomy, University of Hawai‘i at Mānoa • Advisor: Rohan Naidu

July 2026 - present

- Fit stellar atmosphere models to little red dot central engines (black hole stars) and estimated their masses with four non-virial methods, finding masses consistent with black hole stars arising from single supermassive stars.

Decomposing Little Red Dots

Cambridge, MA

MIT Kavli Institute • Advisor: Rohan Naidu

July 2025 - May 2026

- Decomposed 90+ little red dots from the DAWN JWST Archive into black hole star and host galaxy components, finding population-level evidence that every little red dot hosts a black hole star.

Fractal Generative Models

Cambridge, MA

MIT CSAIL, He Vision Group • Advisors: Tianhong Li, Kaiming He

Nov. 2024 - Feb. 2025

- Contributed to building and validating fractal generative models that abstract generative models into atomic generative modules.

DafnyBench: A Benchmark for Formal Software Verification

Cambridge, MA

NSF IAIFI, Tegmark AI Safety Group • Advisor: Max Tegmark

Feb. - Nov. 2024

- Built [DafnyBench](#), the largest formal software verification benchmark at the time (750+ programs), and evaluated large language models’ capability to write formally verifiable code.
- Built [Dafny Autopilot](#), a VS Code extension that leverages GPT-4o or Claude 3.5 Sonnet to verify AI-generated code (400+ installs).

Focused Skill Discovery: Using Per-Factor Empowerment to Control State Variables

Berkeley, CA

Center for Human-Compatible AI, UC Berkeley • Advisor: Cam Allen

June - Aug. 2024

- Investigated focused skill discovery in reinforcement learning, where agents learn skills that change one state variable at a time to explore more efficiently and avoid side effects.

RELEVANT COURSEWORK

Physics	Astrophysics I (grad), General Relativity (grad), Statistical Mechanics I (grad), Relativistic Quantum Field Theory I (grad), Electromagnetic Theory (grad)
Mathematics	Real Analysis, Numerical Analysis, Theory of Computation, Combinatorial Optimization